

CTFs (Capture the Flag)

HackTheBox

TryHackMe

VulnHub

picoCTF

pwn.college

Certifications

Beginner Certifications

CompTIA A+

CompTIA Linux+

CompTIA Network+

CCNA

CompTIA Security+

Advanced Certifications

CEH

CISA

CISM

GSEC

GPEN

GWAPT

GIAC

OSCP

CREST

CISSP

Common Virtualization Technologies

VMWare

VirtualBox

esxi

proxmox

Basics of Virtualization

Hypervisor

GuestOS

HostOS

VM

ipconfig

ping

dig

netstat

route

nmap

tcpdump

arp

tracert

nslookup

iptables

Packet Sniffers

Port Scanners

Protocol Analyzers

Troubleshooting Tools

Kerberos

RADIUS

LDAP

SSO

Certificates

Local Auth

Authentication Methodologies

Cyber Security

Fundamental IT Skills

Computer Hardware Components

Connection Types and their function

OS-Independent Troubleshooting

Understand Basics of Popular Suites

Basics of Computer Networking

Basics of Subnetting

Public vs Private IP Addresses

IP Terminology

localhost

loopback

CIDR

subnet mask

default gateway

Understand the Terminology

VLAN

DMZ

ARP

VM

DHCP

DNS

NAT

IP

Router

Switch

VPN

MAN

LAN

WAN

WLAN

Understand these

DHCP

DNS

NTP

IPAM

Functions of each

Star

Ring

Mesh

Bus

Network Topologies

SSH

RDP

FTP

SFTP

HTTP / HTTPS

SSL / TLS

Network Protocols

Find the detailed version of this roadmap along with other similar roadmaps

roadmap.sh

NFC

Bluetooth

WiFi

Infrared

MS Office Suite

iCloud

Google Suite

Windows

Linux

MacOS

Operating Systems

Learn following for each

Installation and Configuration

Different Versions and Differences

Navigating using GUI and CLI

Understand Permissions

Installing Software and Applications

Performing CRUD on Files

Troubleshooting

Common Commands

Networking Knowledge

Understand the OSI Model

Common Protocols and their Uses

Common Ports and their Uses

SSL and TLS Basics

Basics of NAS and SAN

Security Skills and Knowledge

Understand Common Hacking Tools

Core Concepts of Zero Trust

Understand Common Exploit Frameworks

Roles of Compliance and Auditors

Understand Concept of Defense in Depth

Understand the Definition of Risk

Understand Concept of Runbooks

Understand Backups and Resiliency

Understand Basics of Forensics

Cyber Kill Chain

Basics and Concepts of Threat Hunting

MFA & 2FA

Honeypots

Basics of Vulnerability Management

Operating System Hardening

Basics of Reverse Engineering

Understand Concept of Isolation

Penetration Testing Rules of Engagement

Basics of IDS and IPS

Perimeter vs DMZ vs Segmentation

Authentication vs Authorization

Blue / Red / Purple Teams

False Negative / False Positive

True Negative / True Positive

Basics of Threat Intel, OSINT

Understand Handshakes

Understand CIA Triad

Privilege Escalation

Web Based Attacks and OWASP10

Learn how Malware works and Types

Tools for Incident Response and Discovery

dig

nmap

ping

arp

cat

dd

tail

hping

head

grep

nslookup

tracert

winhex

autopsy

ipconfig

curl

wireshark

memdump

FTK Imager

Understand Common Standards

ISO

RMF

NIST

CIS

CSF

Using tools for Unintended Purposes

LOLBAS

GTFEBINS

WADCOMS

Learn how to find and use these logs

Event Logs

syslogs

netflow

Packet Captures

Firewall Logs

Understand Hardening Concepts

MAC-based

NAC-based

Port Blocking

Group Policy

Sinkholes

ACLs

Patching

Jump Server

Endpoint Security

Understand Common Tools

VirusTotal

urlscan

any.run

Joe Sandbox

urlvoid

WHOIS

Understand Audience

Stakeholders

HR

Legal

Compliance

Management

Basics of Cryptography

Salting

Hashing

Key Exchange

Private vs Public Keys

PKI

Obfuscation

Understand Frameworks

Diamond Model

Kill Chain

ATT&CK

Common Distro's for hacking

ParrotOS

Kali Linux

Understand the following

SIEM

SOAR

Secure vs Unsecure Protocols

FTP vs SFTP

SSL vs TLS

IPSEC

DNSSEC

LDAPS

SRTTP

S/MIME

Understand the following Terms

Antivirus

Antimalware

EDR

DLP

Firewall & Nextgen Firewall

HIPS

NIDS

NIPS

Host Based Firewall

Sandboxing

EAP vs PEAP

WPS

ACL

WPA vs WPA2 vs WPA3 vs WEP

Understand the Incident Response Process

Preparation

Identification

Containment

Eradication

Recovery

Lessons Learned

Understand Threat Classification

Zero Day

Known vs Unknown

APT

Attack Types and Differences

Phishing

Whishing

Whaling

Smishing

Spam vs Spim

Shoulder Surfing

Tailgating

Dumpster Diving

Zero day

Social Engineering

Reconnaissance

Impersonation

Watering Hole Attack

Drive by Attack

Typo Squatting

Brute Force vs Password Spray

Common Attacks

DoS vs DDoS

MITM

CSRF

Spoofing

SQL Injection

XSS

Evil Twin

VLAN Hopping

DNS Poisoning

Deauth Attack

Replay Attack

Rogue Access Point

Buffer Overflow

Memory Leak

Pass the Hash

Directory Traversal

Cloud Skills and Knowledge

Understand the Concept of Security in the Cloud

Understand the differences between cloud and on-premises

Understand the concept of Infrastructure as Code

Understand the Concept of Serverless

Understand the basics and general flow of deploying in the cloud

Understand Cloud Services

SaaS

PaaS

IaaS

Cloud Models

Private

Public

Hybrid

Common Cloud Environments

AWS

GCP

Azure

Common Cloud Storage

S3

Dropbox

iCloud

Box

OneDrive

Google Drive

Programming Skills

Python

Go

JavaScript

C++

Bash

Power Shell

Keep Learning